八十三年第一次專門職業及技術人員 律師、會計師 檢覈筆試試題 09-4 全一張（正面）
建築師、技師

類　　科：冷凍空調工程技師
科　　目：冷 凍 工 程
考試時間：二 小 時

一、試簡述壓縮式冷凍系統與吸收式冷凍系統在原理上，機構上，有何異同之處。（15分）

二、冷凍系統在選用馬達時，一般要注意那些條件。（15分）

三、家庭用的冰櫃，常令使用者發生困擾有那些？請簡述其發生的原因。（20分）

四、一4極，60赫茲的交流馬達，若帶動壓縮機的實際轉速爲1725RPM，試求其皮帶輪的滑動百分率。併說明滑動率的大小，對冷凍系統有何不良的影響。（20分）

五、一理想空氣標準的冷凍循環，壓縮機的壓縮比爲4:1，空氣進入壓縮機的壓力與溫度分別爲101.3kpa,37°C，出透平機的溫度爲-3°C。
試求：㈠在熱交換器中，每一單位質量的空氣傳遞的熱量。（10分）
㈡每一單位質量空氣在壓縮機及透平機中作的功。（10分）
㈢此冷凍系統的冷凍效應係數(COP)。（10分）

參考附表：

Table B-1 SATURATED WATER—TEMPERATURE TABLE

TEMP., °C, T	PRESS., kPa P	SPECIFIC VOLUME, m³/kg SAT. LIQUID, v_f	SAT. VAPOR, v_g	INTERNAL ENERGY, kJ/kg SAT. LIQUID, u_f	EVAP., u_{fg}	SAT. VAPOR, u_g	ENTHALPY, kJ/kg SAT. LIQUID, h_f	EVAP., h_{fg}	SAT. VAPOR, h_g	ENTROPY, kJ/kg·K SAT. LIQUID, s_f	EVAP., s_{fg}	SAT. VAPOR, s_g
0.01	0.6113	0.001 000	206.14	.00	2375.3	2375.3	.01	2501.3	2501.4	.0000	9.1562	9.1562
5	0.8721	0.001 000	147.12	20.97	2361.3	2382.3	20.98	2489.6	2510.6	.0761	8.9496	9.0257
10	1.2276	0.001 000	106.38	42.00	2347.2	2389.2	42.01	2477.7	2519.8	.1510	8.7498	8.9008
15	1.7051	0.001 001	77.93	62.99	2333.1	2396.1	62.99	2465.9	2528.9	.2245	8.5569	8.7814
20	2.339	0.001 002	57.79	83.95	2319.0	2402.9	83.96	2454.1	2538.1	.2966	8.3706	8.6672
25	3.169	0.001 003	43.36	104.88	2304.9	2409.8	104.89	2442.3	2547.2	.3674	8.1905	8.5580
30	4.246	0.001 004	32.89	125.78	2290.8	2416.6	125.79	2430.5	2556.3	.4369	8.0164	8.4533
35	5.628	0.001 006	25.22	146.67	2276.7	2423.4	146.68	2418.6	2565.3	.5053	7.8478	8.3531
40	7.384	0.001 008	19.52	167.56	2262.6	2430.1	167.57	2406.7	2574.3	.5725	7.6845	8.2570
45	9.593	0.001 010	15.26	188.44	2248.4	2436.8	188.45	2394.8	2583.2	.6387	7.5261	8.1648
50	12.349	0.001 012	12.03	209.32	2234.2	2443.5	209.33	2382.7	2592.1	.7038	7.3725	8.0763
55	15.758	0.001 015	9.568	230.21	2219.9	2450.1	230.23	2370.7	2600.9	.7679	7.2234	7.9913
60	19.940	0.001 017	7.671	251.11	2205.5	2456.6	251.13	2358.5	2609.6	.8312	7.0784	7.9096
65	25.03	0.001 020	6.197	272.02	2191.1	2463.1	272.06	2346.2	2618.3	.8935	6.9375	7.8310
70	31.19	0.001 023	5.042	292.95	2176.6	2469.6	292.98	2333.8	2626.8	.9549	6.8004	7.7553
75	38.58	0.001 026	4.131	313.90	2162.0	2475.9	313.93	2321.4	2635.3	1.0155	6.6669	7.6824
80	47.39	0.001 029	3.407	334.86	2147.4	2482.2	334.91	2308.8	2643.7	1.0753	6.5369	7.6122
85	57.83	0.001 033	2.828	355.84	2132.6	2488.4	355.90	2296.0	2651.9	1.1343	6.4102	7.5445
90	70.14	0.001 036	2.361	376.85	2117.7	2494.5	376.92	2283.2	2660.1	1.1925	6.2866	7.4791
95	84.55	0.001 040	1.982	397.88	2102.7	2500.6	397.96	2270.2	2668.1	1.2500	6.1659	7.4159

TEMP., °C, T	PRESS., MPa P	SPECIFIC VOLUME, m³/kg SAT. LIQUID, v_f	SAT. VAPOR, v_g	INTERNAL ENERGY, kJ/kg SAT. LIQUID, u_f	EVAP., u_{fg}	SAT. VAPOR, u_g	ENTHALPY, kJ/kg SAT. LIQUID, h_f	EVAP., h_{fg}	SAT. VAPOR, h_g	ENTROPY, kJ/kg·K SAT. LIQUID, s_f	EVAP., s_{fg}	SAT. VAPOR, s_g
100	0.101 35	0.001 044	1.6729	418.94	2087.6	2506.5	419.04	2257.0	2676.1	1.3069	6.0480	7.3549
105	0.120 82	0.001 048	1.4194	440.02	2072.3	2512.4	440.15	2243.7	2683.8	1.3630	5.9328	7.2958
110	0.143 27	0.001 052	1.2102	461.14	2057.0	2518.1	461.30	2230.2	2691.5	1.4185	5.8202	7.2387
115	0.169 06	0.001 056	1.0366	482.30	2041.4	2523.7	482.48	2216.5	2699.0	1.4734	5.7100	7.1833
120	0.198 53	0.001 060	0.8919	503.50	2025.8	2529.3	503.71	2202.6	2706.3	1.5276	5.6020	7.1296
125	0.2321	0.001 065	0.7706	524.74	2009.9	2534.6	524.99	2188.5	2713.5	1.5813	5.4962	7.0775
130	0.2701	0.001 070	0.6685	546.02	1993.9	2539.9	546.31	2174.2	2720.5	1.6344	5.3925	7.0269
135	0.3130	0.001 075	0.5822	567.35	1977.7	2545.0	567.69	2159.6	2727.3	1.6870	5.2907	6.9777
140	0.3613	0.001 080	0.5089	588.74	1961.3	2550.0	589.13	2144.7	2733.9	1.7391	5.1908	6.9299
145	0.4154	0.001 085	0.4463	610.18	1944.7	2554.9	610.63	2129.6	2740.3	1.7907	5.0926	6.8833

（請接背面）

類　　科：冷凍空調工程技師
科　　目：冷 凍 工 程

Table D-9 ZERO-PRESSURE SPECIFIC HEATS AND k AT 27°C FOR COMMON GASES

GAS	CHEMICAL FORMULA	MOLECULAR WEIGHT	R, kJ/kg·K	c_{p0}, kJ/kg·K	c_{v0}, kJ/kg·K	k
Air	—	28.97	0.287 00	1.005	0.718	1.400
Argon	Ar	39.948	0.208 13	0.5203	0.3122	1.667
Butane	C_4H_{10}	58.124	0.143 04	1.7164	1.5734	1.091
Carbon dioxide	CO_2	44.011	0.188 92	0.846	0.657	1.289
Carbon monoxide	CO	28.011	0.296 83	1.040	0.744	1.400
Ethane	C_2H_6	30.070	0.276 50	1.7662	1.4897	1.186
Ethylene	C_2H_4	28.054	0.296 37	1.5482	1.2518	1.237
Helium	He	4.003	2.077 03	5.1926	3.1156	1.667
Hydrogen	H_2	2.016	4.124 18	14.307	10.183	1.405
Methane	CH_4	16.043	0.518 35	2.2537	1.7354	1.299
Neon	Ne	20.183	0.411 95	1.0299	0.6179	1.667
Nitrogen	N_2	28.013	0.296 80	1.039	0.743	1.400
Octane	C_8H_{18}	114.23	0.072 79	1.7113	1.6385	1.044
Oxygen	O_2	31.999	0.259 83	0.918	0.658	1.395
Propane	C_3H_8	44.097	0.188 55	1.6794	1.4909	1.126
Steam	H_2O	18.015	0.461 52	1.8723	1.4108	1.327

SOURCE: Gordon J. Van Wylen and Richard E. Sonntag, *Fundamentals of Classical Thermodynamics*, SI Version, 2nd Ed, Revised Printing, Copyright © 1976, 1978 by John Wiley & Sons, Inc. Table A-8, pp. 683.

Table D-1 IDEAL-GAS PROPERTIES OF AIR (MASS BASIS)

T, K	h, kJ/kg	u, kJ/kg	s°, kJ/kg·K	P_r	v_r
100	99.76	71.06	5.6008	0.02990	2230
110	109.77	78.20	5.6963	0.04171	1758.4
120	119.79	85.34	5.7836	0.05652	1415.7
130	129.81	92.51	5.8638	0.07474	1159.8
140	139.84	99.67	5.9380	0.09681	964.2
150	149.86	106.81	6.0071	0.12318	812.0
160	159.87	113.95	6.0718	0.15431	691.4
170	169.89	121.11	6.1326	0.19068	594.5
180	179.92	128.28	6.1898	0.23279	515.6
190	189.94	135.40	6.2440	0.28114	450.6
200	199.96	142.56	6.2953	0.3363	396.6
210	209.97	149.70	6.3442	0.3987	351.2
220	219.99	156.84	6.3908	0.4690	312.8
230	230.01	163.98	6.4354	0.5477	280.0
240	240.03	171.15	6.4780	0.6355	251.8
250	250.05	178.29	6.5190	0.7329	227.45
260	260.09	185.45	6.5582	0.8405	206.26
270	270.12	192.59	6.5961	0.9590	187.74
280	280.14	199.78	6.6326	1.0889	171.45
290	290.17	206.92	6.6678	1.2311	157.07
300	300.19	214.09	6.7018	1.3860	144.32
310	310.24	221.27	6.7348	1.5546	132.96
320	320.29	228.45	6.7667	1.7375	122.81
330	330.34	235.65	6.7976	1.9352	113.70
340	340.43	242.86	6.8277	2.149	105.51
350	350.48	250.05	6.8569	2.379	98.11
360	360.58	257.23	6.8852	2.626	91.40
370	370.67	264.47	6.9129	2.892	85.31
380	380.77	271.72	6.9399	3.176	79.77
390	390.88	278.96	6.9661	3.481	74.71
400	400.98	286.19	6.9917	3.806	70.07
410	411.12	293.45	7.0167	4.153	65.83
420	421.26	300.73	7.0412	4.522	61.93
430	431.43	308.03	7.0651	4.915	58.34
440	441.61	315.34	7.0885	5.332	55.02
450	451.83	322.66	7.1114	5.775	51.96
460	462.01	329.99	7.1338	6.245	49.11
470	472.25	337.34	7.1558	6.742	46.48
480	482.48	344.74	7.1774	7.268	44.04
490	492.74	352.11	7.1985	7.824	41.76
500	503.02	359.53	7.2193	8.411	39.64
510	513.32	366.97	7.2397	9.031	37.65
520	523.63	374.39	7.2598	9.684	35.80
530	533.98	381.88	7.2795	10.372	34.07
540	544.35	389.40	7.2989	11.097	32.45
550	554.75	396.89	7.3179	11.858	30.92

SOURCE: Adapted to SI units from *Gas Tables* by Joseph H. Keenan and Joseph Kaye. Copyright by John Wiley & Sons, Inc., 1948.